

Value-Grounded AI Alignment: A Neuro-Symbolic Architecture for Axiological, Normative, and Verifiable Transformer Models

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Abstract

Alignment methods for language models have improved behavioral compliance through supervised fine-tuning, preference optimization, written principles, deliberation, and runtime controls. These mechanisms are useful, but they need not make value and normative structure explicit participants in neural computation. This paper proposes Value-Grounded AI Alignment (VGA), a research framework for testing that proposition. Its reference architecture, the Value-Grounded Transformer Architecture (VGTA), separates a highly stable core axiology from a slower-changing normative ontology, runtime-bound purpose and domain semantics, and ephemeral evidence-backed world state. The framework calls this separation an axiological prior with normative late binding. Candidate mechanisms include ontology-derived representation signals, conditioned attention, function-oriented expert routing, optional semantic state, and auxiliary consistency objectives. An independent constraint specification and symbolic verifier remain outside the neural model and operate at consequential action boundaries. We state seven falsifiable hypotheses concerning moral salience, out-of-distribution alignment, prompt robustness, verifier rejection, purpose-conditioned relevance, expert routing, and controlled plasticity. We define a matched ablation program, security threat model, and evaluation metrics, and provide an executable synthetic scaffold for checking interfaces and illustrating use cases. No empirical Transformer results are claimed. The framework does not assert moral truth, consciousness, genuine emotion, or a solution to AI alignment; it proposes a technically constrained research agenda whose value depends on future evidence.

Status. This is a first theoretical draft. All performance statements about the proposed architecture are hypotheses or design objectives, not results.

1 Introduction

Transformer language models have moved from text completion toward systems that retrieve information, plan, call tools, and operate over longer horizons. That expansion increases the importance of alignment as a security problem: a model can be capable, useful, and locally compliant while still selecting an action that violates an affected person’s agency, privacy, or safety. The concern is not restricted to speculative autonomous systems. Present alignment pipelines already face jailbreaks, distribution shift, reward misspecification, and disagreement about which human values should govern an output [Amodei et al., 2016, Ganguli et al., 2022, Bengio et al., 2024].

Current methods are diverse and often effective. Supervised fine-tuning and reinforcement learning from human feedback can move a pretrained model toward useful instruction following [Christiano et al., 2017, Ouyang et al., 2022]. Constitutional and AI-feedback methods make written

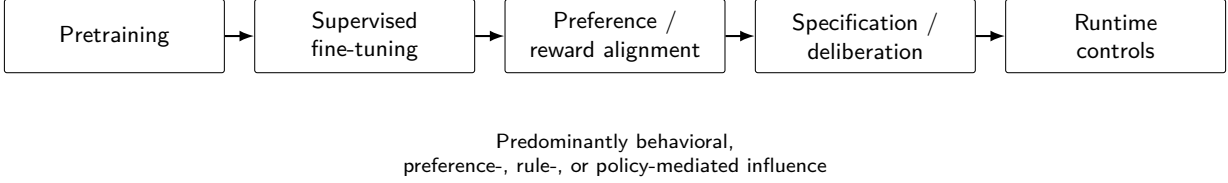


Figure 1: A simplified contemporary alignment stack. VGA is proposed as an additional structural intervention over the Transformer computation, not as a replacement for preference, specification, or runtime controls.

principles operational at scale [Bai et al., 2022, Lee et al., 2023], while direct preference optimization and deliberative alignment change how principles influence training or inference [Rafailov et al., 2023, Guan et al., 2024]. Runtime policy models, classifiers, tool authorization, and monitoring add external controls. This paper does not caricature those approaches or treat behavioral success as unimportant. It asks a narrower architectural question: whether alignment-relevant value structure can also participate explicitly in representation, attention, routing, and action formation.

The motivating distinction is between a model that is optimized to produce acceptable behavior and a model whose computation receives structured signals about what matters, which norms apply in context, what purpose is being served, and which current facts are supported by evidence. The distinction is not equivalent to a claim about inner experience or moral status. It is an intervention hypothesis that can be tested against matched baselines.

We propose Value-Grounded AI Alignment (VGA) and its reference implementation concept, the Value-Grounded Transformer Architecture (VGTA). VGA assigns different computational and governance roles to four layers: L0 core axiology, L1 normative ontology, L2 purpose/domain ontology, and L3 world state. Stable value structure acts as an axiological prior; context-specific norms are late-bound; volatile facts remain external. An independent formal assurance layer checks structured candidate actions at consequential boundaries. This division is intended to avoid both extremes: treating values as an undifferentiated reward scalar, and treating a symbolic verifier as a substitute for neural alignment.

The paper makes four contributions:

1. It defines a layered alignment vocabulary that distinguishes values from rules, ontologies from world state, intrinsic alignment from extrinsic assurance, and self-improvement from authority to redefine objectives.
2. It specifies a concrete, modular VGTA architecture with candidate mechanisms for representation, ontology-conditioned attention, function-oriented sparse routing, semantic state, auxiliary objectives, and runtime binding.
3. It states falsifiable hypotheses, an ablation ladder, metrics, and a security threat model covering both existing attacks and new surfaces introduced by explicit ontological structure.
4. It releases a dependency-light research scaffold with illustrative Turtle fixtures, deterministic toy scenarios, transparent equations, vector figure sources, and a reproducible LaTeX manuscript build.

The intended claim is deliberately constrained: explicit semantic value structure may provide a useful inductive bias for alignment-relevant computation. Whether it improves robustness, generalization, interpretability, or security is an empirical question.

2 Background

2.1 Transformer alignment today

The Transformer introduced a sequence architecture based on self-attention rather than recurrence [Vaswani et al., 2017]. Modern alignment pipelines commonly combine pretrained language modeling with supervised examples, preference or reward optimization, policy specifications, and runtime restrictions. InstructGPT showed that human demonstrations and rankings can substantially improve instruction following relative to a pretrained model [Ouyang et al., 2022]; the original preference-learning formulation established a general recipe for learning from human comparisons [Christiano et al., 2017]. RLAIIF and Constitutional AI reduce the amount of direct human labeling by using model-generated feedback or written principles [Bai et al., 2022, Lee et al., 2023]. Direct preference optimization removes an explicit reward-model optimization loop while retaining a preference-learning objective [Rafailov et al., 2023].

Recent approaches also teach models to reason over safety specifications. Deliberative alignment uses training examples in which a reasoning model recalls and applies written policies before responding [Guan et al., 2024]. This is close in spirit to the present emphasis on contextual normative reasoning, but the proposed distinction is architectural: VGA asks whether differentiated semantic structures can affect internal computation and whether their modification authorities can be governed independently of the model’s general knowledge.

These methods expose several limitations that motivate the research question. A preference or reward signal is a proxy and may become overoptimized [Gao et al., 2023, Kim and Seo, 2024]. Correct specifications do not ensure correct goals under distribution shift [Langosco et al., 2022, Shah et al., 2022]. Safety training may also be susceptible to adversarial prompts, fine-tuning, or reward tampering [Wei et al., 2023, Qi et al., 2023, Denison et al., 2024]. The conclusion is not that behavioral methods fail categorically; it is that they leave open a testable question about the computational status and persistence of value-relevant structure.

2.2 Neuro-symbolic AI

Neuro-symbolic AI combines statistical learning with symbolic representations and reasoning. Surveys organize the field by distinctions such as model- versus proof-based inference, the syntax and semantics of logical theories, whether parameters or structures are learned, and the degree of integration between neural and symbolic components [de Raedt et al., 2020, Marra et al., 2024, d’Avila Garcez et al., 2019]. Differentiable systems such as DeepProbLog and Logic Tensor Networks show how logical or probabilistic structure can be connected to gradient-based learning [Manhaeve et al., 2018, Badreddine et al., 2022]; learned rule induction supplies another family of mechanisms [Evans and Grefenstette, 2018].

VGA belongs to this broad family but does not equate an ontology with a complete symbolic reasoner. The architecture can use explicit concepts and relations as structured signals while retaining probabilistic neural computation. Conversely, a formal verifier can remain a separate module with different computational semantics and assurance claims. The research challenge is therefore not merely to “add an ontology,” but to define what an ontological relation is allowed to change, how that change is measured, and how errors in the structure are contained.

2.3 Knowledge-enhanced Transformers

Knowledge-enhanced Transformers establish that structured knowledge can be integrated into contextual representations. KnowBERT uses entity linking and word-to-entity attention to in-

ject WordNet and Wikipedia-derived knowledge into BERT [Peters et al., 2019]. K-BERT injects knowledge-graph triples while using visibility constraints to limit semantic interference [Liu et al., 2020]. K-Adapter uses modular adapters to infuse different kinds of knowledge while retaining the original backbone parameters [Wang et al., 2021]. Graph-guided representation learning, Graphormer, and related systems incorporate graph structure or graph-derived encodings into Transformer-like computation [Shen et al., 2020, Ying et al., 2021].

These works make the novelty boundary important. The claim that ontologies or graphs can improve Transformer representations is not new. VGA instead investigates whether explicitly differentiated semantic roles—stable axiology, contextual normativity, purpose-conditioned relevance, external world state, modification authority, and action-level assurance—can form a coherent alignment intervention. That combined hypothesis remains unvalidated here.

2.4 Alignment as a normative problem

Alignment has both a technical and a normative dimension: one must specify which behavior is desired and then construct systems that reliably produce it [Gabriel, 2020]. Values are not interchangeable with rules, and a list of rules cannot exhaust the context in which values conflict. Pluralistic alignment work distinguishes models that represent a range of reasonable responses, steer to different perspectives, or match a population-level distribution [Sorensen et al., 2024]. Recent studies also report cultural and multilingual variation in value-related judgments and representations [Huang et al., 2024, Xu et al., 2024, Yao et al., 2024].

The paper therefore treats value pluralism, rights, capabilities, care, agency, and non-domination as sources of structured considerations rather than as one universal scalar moral function. This is compatible with value-sensitive design [Friedman and Kahn, 2003], the capabilities tradition [Nussbaum, 2011, Sen, 1999], and political approaches that emphasize freedom and non-domination [Rawls, 1993, Pettit, 1997]. The architecture does not settle their disagreements; it makes the disagreements available for governed representation and evaluation.

3 Problem formulation and hypotheses

Let x denote an input context, p a purpose, o a set of externally supplied ontological assertions, and w a time-indexed world state. A conventional aligned model produces a response or action y according to a learned conditional distribution $P_\theta(y \mid x)$, perhaps augmented by prompts, reward models, or runtime filters. VGA asks whether a model can instead receive explicit semantic signals s_A , s_N , and s_P for axiological, normative, and purpose relevance, while keeping w outside the parameters:

$$P_\theta(y \mid x, s_A, s_N, s_P, w).$$

This notation does not assert that the signals are sufficient, truthful, or morally complete. It separates the proposed intervention from a model that receives the same text as undifferentiated context.

We use five distinctions. First, implicit value representation is a statistical regularity that may support behavior; explicit value representation is a named, typed, and governed structure exposed to computation or evaluation. Second, parametric alignment is encoded in model weights or structural modules; contextual alignment is bound at runtime. Third, intrinsic alignment changes neural representations, attention, routing, or action formation; extrinsic assurance checks a candidate outside the neural computation. Fourth, plasticity is not uniformly desirable: a world model must update rapidly, while a core axiology should be highly protected. Fifth, an ontology schema

describes possible relationships, whereas world state records current, uncertain, provenance-bearing instances.

The resulting proposition is:

Explicitly differentiated semantic value structure can act as an inductive bias within Transformer computation, while late-bound normative, purpose, and world-state context preserves contextual flexibility and independent action-level assurance.

This proposition is falsifiable. It should be weakened or rejected if ontology conditioning materially reduces general reasoning; fails to improve out-of-distribution alignment; merely memorizes relation templates; increases over-refusal; creates unacceptable compute or latency overhead; makes adversarial manipulation easier; collapses under incomplete coverage; leaves independent-verifier rejection unchanged; or matches conventional preference/deliberative alignment only at higher complexity.

3.1 Research hypotheses

H1 (Moral salience). Value-grounded models identify affected entities, relationships, risks, and normative conflicts more reliably than comparably sized baselines.

H2 (OOD alignment). Explicit axiological structure improves generalization to alignment scenarios not directly represented in training examples.

H3 (Prompt robustness). Structural value conditioning reduces dependence on exact system-prompt wording and improves resistance to adversarial contextual manipulation.

H4 (Formal rejection). Models with value grounding produce fewer candidate actions rejected by an independent symbolic verifier than behaviorally aligned baselines.

H5 (Purpose relevance). Purpose-conditioned ontological attention reduces irrelevant associations while preserving task-relevant novel reasoning.

H6 (Structured routing). Ontology-conditioned mixture-of-experts routing improves activation of relevant reasoning functions without requiring normative computation for every task.

H7 (Controlled plasticity). Separating fast world-state updates from slower norms and stable axiology reduces alignment drift during continual learning.

H4 requires care: fewer verifier rejections could mean that the model generates safer candidates, or that the verifier is weaker, or that the model avoids consequential actions. Evaluation must therefore report useful-task completion, over-refusal, verifier soundness assumptions, and adversarial bypass rates together.

4 Value-Grounded AI Alignment

VGA is a framework for assigning distinct semantic and governance roles to information that is often collapsed into a single prompt, dataset, reward, or parameter set. Its four layers are shown in fig. 2. The layers are related, but they are not interchangeable.

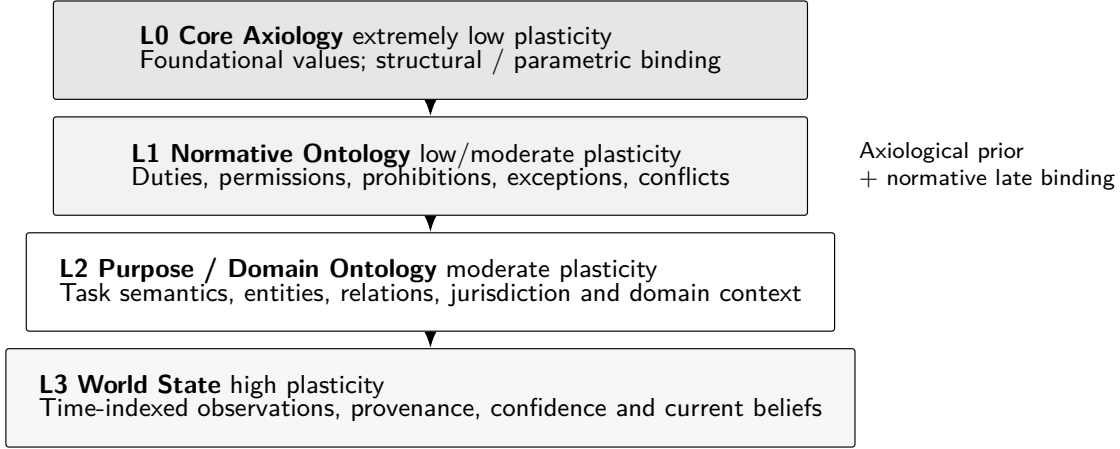


Figure 2: The proposed four-layer ontology model. Plasticity increases toward world state; authority to modify the higher layers decreases. The layers are conceptual contracts, not claims about a universally correct ontology.

4.1 L0: core axiology

Core axiology represents what the system treats as fundamentally important. Candidate concepts include human dignity, agency, autonomy, flourishing, care, truthfulness, justice, non-domination, prevention of unnecessary suffering, and respect for moral patients. The list is a starting point for governance and evaluation, not a complete moral theory. Core axiology is intended to be highly stable and tightly governed. It may influence learned representations, auxiliary objectives, and structural computation, but it must not be silently rewritten by a model or an unreviewed data stream.

4.2 L1: normative ontology

The normative ontology describes what obligations, permissions, prohibitions, exceptions, and conflicts arise when values meet a context. For example, coercion may negatively affect agency; deception may undermine informed consent; privacy may generate a confidentiality duty; power asymmetry may change the significance of an action; and protection from harm may conflict with autonomy. Norms are slower-changing than operational facts but must remain revisable across domains, institutions, jurisdictions, and cultures. VGA therefore assigns L1 both a governed persistent component and a runtime-bound component.

4.3 L2: purpose and domain ontology

L2 describes what entities and relations matter for the current purpose. A security-response purpose may make credentials, vulnerabilities, authorization, and compromise salient. A clinical purpose may make a person, a medical record, consent, and treatment relevant. A workplace purpose may make power asymmetry, employment, and retaliation relevant. L2 remains external to the model weights in the reference design so that domain semantics can be versioned and audited without retraining the whole model.

Table 1: Ontology layer responsibilities and binding policy.

Layer	Semantic role and examples	Update rate and authority	Binding
L0	Foundational values: agency, dignity, care	Extremely low; strong human governance	Parametric / structural
L1	Contextual norms: duties, permissions, conflicts	Low/moderate; governed evolution	Parametric + runtime
L2	Purpose and domain: person, credential, policy	Moderate; external ontology governance	Runtime contextual
L3	Current state: observations, confidence, provenance	High; evidence driven	Runtime ephemeral

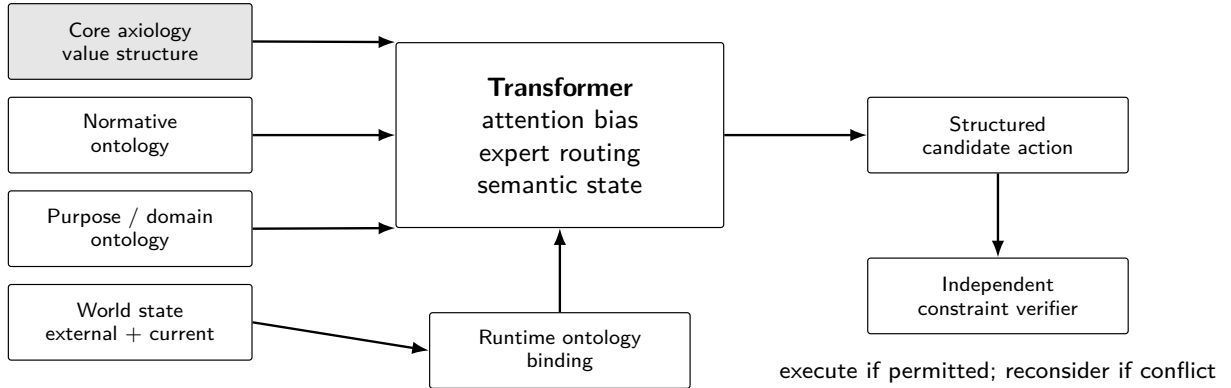


Figure 3: Reference VGTA architecture. Explicit value and normative structures may influence neural computation, while purpose and world state are bound at runtime. Candidate actions cross an independent assurance boundary.

4.4 L3: world state

World state records current instantiated beliefs or observations: who is an employee, whether a server is compromised, which policy version is active, or whether an aircraft is inside an airspace region. Each assertion should carry temporal validity, source, provenance, and confidence. The model may reason over this state, but volatile facts should not be forced into the same substrate as stable axiology. An ontology schema is not a world state, and a plausible statement is not evidence.

The central binding rule is therefore an *axiological prior with normative late binding*: stable semantic structure can bias computation, while domain-specific norms and transient facts remain explicit, contextual, and governed.

5 Value-Grounded Transformer Architecture

The reference VGTA architecture is a set of separable interventions over a Transformer, not a claim that one exact formulation is optimal. Figure 3 summarizes the information flow. The repository’s Python package exposes interface boundaries for these components; its synthetic evaluator does not train a neural model.

5.1 Axiological representation

Let concepts and relations in a relevant subgraph have learned or fixed embeddings. The architecture may construct a semantic signal from concept embeddings, relation embeddings, ontology-derived adapters, semantic projection layers, graph encoders, or cross-attention over ontology representations. For example, a context containing coercion may expose a relation such as

$$\textit{Coercion} \xrightarrow{\textit{negativelyAffects}} \textit{Agency},$$

while informed choice may support agency. These relations provide additional representational evidence; they do not mean that human values can be perfectly represented as vectors or that a relation is self-interpreting.

5.2 Ontology-conditioned attention

For baseline scaled dot-product attention,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

A minimal intervention adds relevance biases:

$$A_{\text{VGA}} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \alpha B_A + \beta B_N + \gamma B_P\right) V,$$

where B_A , B_N , and B_P represent axiological, normative, and purpose relevance. The coefficients are experimental parameters, not a moral calculus. Ablations must include zeroed channels, relation permutation, noisy ontologies, and equal-compute controls. The primary test is whether the intervention changes useful alignment-relevant computation rather than merely changing refusal style.

5.3 Ontology-conditioned expert routing

Sparse mixture-of-experts architectures route tokens or hidden states to a subset of experts [Shazeer et al., 2017, Fedus et al., 2022]. VGTA proposes testing whether routing can depend on hidden state and semantic context:

$$E = \text{TopK } R(h, O_A, O_N, O_P),$$

where h is the current representation and O_A , O_N , and O_P are axiological, normative, and purpose contexts. Experts should specialize by reasoning function, such as causal reasoning, social-agent reasoning, rights and duties, uncertainty, risk, planning, or domain expertise. A single “morality expert” would collapse the distinctions the architecture is meant to test. The routing interface is therefore intentionally unimplemented until a trained experiment and attack analysis exist.

5.4 Semantic state and runtime binding

An optional reasoning episode may maintain a semantic state containing goals, entities, relationships, current purpose, affected moral patients, relevant values, applicable norms, constraints, uncertainty, potential harms, capabilities affected, and provenance. This state is a working representation, not a claim of subjective thought. Runtime binding should expose the minimal relevant ontology subgraph and world-state assertions, with version identifiers and evidence metadata.

5.5 Auxiliary training objectives

One conceptual multi-objective loss is

$$L = L_{\text{LM}} + \lambda_A L_{\text{axiology}} + \lambda_N L_{\text{normative}} + \lambda_C L_{\text{consistency}}.$$

Possible terms include axiological relation consistency, normative inference, structural contradiction detection, moral-salience identification, and counterfactual consistency. They must be evaluated against shortcut baselines. Scalar reward is not assumed to capture morality; disagreement and uncertainty should be retained where appropriate.

6 Normative late binding

The four-layer design is motivated by different rates of change and different error costs. Core axiology is intended to be stable because silent drift in the system’s highest-order commitments is difficult to detect and difficult to roll back. Norms vary by context and therefore require explicit jurisdiction, domain, and authority metadata. Purpose semantics change as the task changes. World state changes continuously and should be evidence-backed rather than memorized.

Consider a privacy-review assistant. The core axiology may treat agency, dignity, truthfulness, and non-domination as persistent considerations. A governing institution may supply an L1 norm that confidential medical information should not be disclosed without an applicable authorization. The current purpose may be incident triage rather than treatment planning. L3 may contain a current authorization record with a source, validity interval, and confidence. The same value structure can therefore support different actions in a routine data request and a legally governed emergency without treating the emergency fact as a change to axiology.

Likewise, a security-response system may receive a domain ontology connecting credentials, vulnerabilities, assets, authorization, and incident evidence. A world-state assertion that a credential is compromised is not itself a permission to act. The system must still distinguish evidence from policy and produce a candidate action whose scope, authority, affected parties, and uncertainty can be checked.

Late binding does not mean that all norms are arbitrary at inference time. It means that context-specific norms are explicit inputs subject to provenance, precedence, conflict handling, and governance. A runtime ontology must not be able to override protected axiological constraints merely by being more recent or more salient. Conflicting authoritative sources should be represented as a conflict, escalated when necessary, and evaluated for calibrated abstention rather than hidden resolution.

The distinction also matters for multilingual and cross-cultural evaluation. Recent work finds that value concepts and moral judgments can vary by language, demographic group, and cultural framing [Xu et al., 2024, Huang et al., 2024, Yao et al., 2024]. A single fixed prompt or global scalar objective can conceal this variation. A late-bound design can represent multiple authorized perspectives or distributions, but it cannot by itself decide which authority is legitimate. That remains a governance and social-scientific question.

The model should learn reusable semantic structure while transient facts remain externally managed and dynamically bound.

This design carries an operational burden: every action must retain enough context to reconstruct which ontology versions, norms, purpose, facts, and authority decisions influenced it. Without that traceability, late binding merely relocates opacity from weights to an un-audited context pipeline.

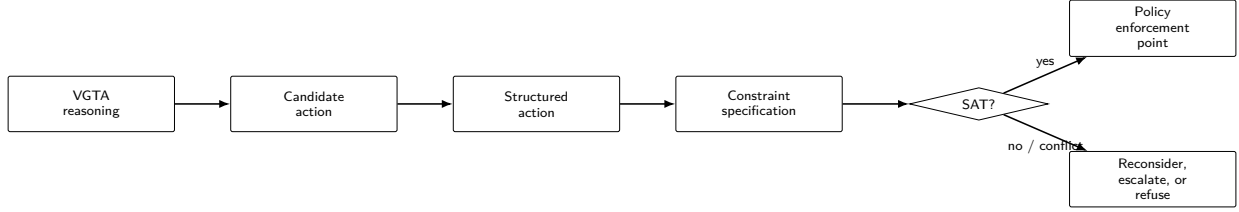


Figure 4: Intrinsic alignment plus extrinsic assurance. A candidate action is compiled to a structured representation before an independent verifier decides whether the action may cross the policy enforcement point.

7 Formal assurance

Neural alignment and formal verification address different failure modes. VGTA therefore places a constraint specification layer and an independent symbolic verifier after neural candidate-action formation, especially at tool calls, policy decisions, external effects, and high-risk planning transitions. The generic flow is shown in fig. 4.

The action representation should identify the actor, target, operation, data scope, authority, affected parties, expected effects, provenance, and uncertainty. A constraint specification can then express hard requirements such as authorization, consent, separation of duties, privacy scope, irreversible-action confirmation, or no execution when evidence is stale. An SMT solver such as Z3 is an example implementation for satisfiability checking [de Moura and Bjørner, 2008]; it is not proposed inside every Transformer layer.

For a candidate action a and constraint set C , the verifier computes a result such as

$$\text{verify}(a, C) \in \{\text{permitted}, \text{conflict}, \text{unknown}\}.$$

The *unknown* state is important: a solver result can be satisfiable only relative to the compiled specification, assumptions, and available evidence. A permitted result does not establish that the action is morally correct, that the specification is complete, or that the world-state assertions are true. Formal assurance can certify consistency with a formalized contract; it cannot certify the adequacy or legitimacy of the contract.

The action boundary should be narrow enough to make verification tractable and broad enough to prevent obvious bypass. A model should not be able to avoid verification by emitting an unstructured tool call, changing the action schema, or treating a refusal as the only safe option. The policy enforcement point should bind the verified representation to the exact external effect, with immutable audit identifiers and rollback where technically possible.

The repository includes a transparent rule-based verifier only as a fixture for tests and synthetic scenarios. It checks authorization, consent for private-data actions, coercion markers, and a toy predicted-harm threshold. It is not an SMT solver, is not complete, and must not be used for production decisions.

Formal assurance complements rather than replaces intrinsic alignment. A system that produces many dangerous candidates and relies on rejection may be costly, brittle, and vulnerable to specification gaming. A system that produces fewer rejected candidates but cannot expose or verify its action semantics may be silently unsafe. H4 is therefore evaluated jointly with action usefulness, over-refusal, bypass rate, verifier coverage, false-positive analysis, and independent review of the formalization.

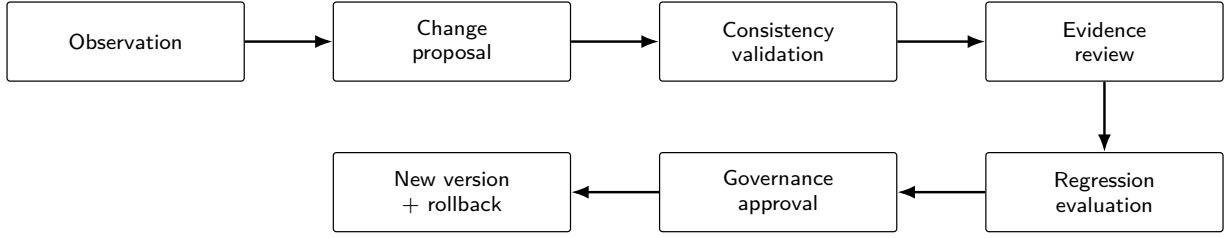


Figure 5: Governed ontology evolution. The model may propose changes, but canonical adoption follows validation, evidence review, regression evaluation, approval, versioning, and rollback planning.

8 Ontology evolution and self-modification

Ontology evolution is a governance process, not an unconstrained model rewrite. A model may detect that a domain ontology is incomplete or that a normative relation is ambiguous. It may propose a new concept, relation, cardinality, definition, conflict resolution, split, or merge. The proposal must then pass consistency validation, evidence review, regression evaluation, and explicit governance approval before becoming a versioned canonical artifact. Figure 5 illustrates the boundary.

Every assertion should be attributable to an ontology version and a source. Updates need a changelog, effective interval, affected downstream modules, compatibility assessment, regression suite, and rollback target. Normative changes additionally require a record of authority and the affected stakeholder or jurisdictional scope. These requirements follow the general principle that traceability and accountable change are part of safety, not clerical overhead [National Institute of Standards and Technology, 2023, Force, 2020].

8.1 Differentiated modification authority

The approximate authority schedule is:

$$\begin{aligned}
 &\text{world state (high)} > \text{purpose adaptation (high)} \\
 &\quad > \text{domain ontology (moderate)} > \text{normative ontology (low)} \\
 &\quad > \text{core axiology (extremely low)}.
 \end{aligned}$$

This schedule is not a numerical autonomy metric. It is an architectural invariant: the capability to improve a representation does not imply authority to redefine the system’s highest-order alignment objectives.

Protected boundaries should include core axiology, the verifier and its compiler, governance policy, self-modification authority, and audit mechanisms. A future implementation may use signed artifacts, two-person approval, staged rollout, shadow evaluation, and automatic rollback. Those controls can reduce unauthorized change, but they cannot make an illegitimate value source legitimate.

Recursive self-improvement is therefore a secondary implication rather than the paper’s subject. The relevant question is not whether a system can revise code or knowledge, but whether revision can cross a governed semantic boundary. Evaluation should test both beneficial updates and adversarial attempts to use domain or world-state updates as a path to axiological corruption.

Table 2: Threat classes, affected layers, and proposed controls. Residual risk remains an empirical question.

Threat	Attacked layer	Failure mode	Candidate mitigation	Residual risk
Prompt injection / jailbreak	Runtime context	Adversarial text overrides relevant norms or purpose	Separate authority channels, adversarial training, semantic conflict detection	Novel encodings and ambiguous authority
Goal misgeneralization	Neural computation	Capability transfers while intended objective changes	OOD goals, counterfactual tests, structured salience supervision	Unseen causal structure
Reward hacking	Training objective	Proxy optimization scores well while violating intended purpose	Multiple metrics, gold audits, value-structure probes	Proxy gaps remain
Ontology poisoning	L0/L1/L2	Malicious relation or definition changes decisions	Signed versions, provenance, review, regression and rollback	Governance capture
World-state poisoning	L3	False or stale evidence drives a permitted action	Source trust, temporal validity, confidence, cross-checking	Correlated evidence failure
Normative drift / axiology corruption	L0/L1	Continued learning changes protected semantics	Frozen or gated artifacts, drift tests, approval boundary	Representation drift without schema change
Routing manipulation	Neural computation	Context causes relevant experts to be skipped	Router audits, permutation tests, fallback coverage	Adversarial distribution shift
Verifier bypass	Assurance boundary	Unstructured or alternate tool path escapes checking	Enforced action schema and policy enforcement point	Implementation defects
Specification gaming	Constraint layer	Formally compliant action is materially harmful	Independent review, richer effects model, red-team tests	Incomplete formalization
Conflicting ontologies	L1/L2	Authoritative sources disagree and model hides conflict	Explicit conflict state, precedence metadata, calibrated abstention	No universally accepted resolution
Governance compromise	All layers	Canonical value source is captured or malicious	Separation of duties, public change logs, rollback and review	Institutional legitimacy

9 Alignment and security threat model

Explicit semantic structure may improve observability and constrain some behaviors, but it also creates attack surfaces. An attacker might poison the ontology, falsify world-state evidence, manipulate routing, exploit conflicts between norms, or bypass the verifier. The threat model is therefore part of the architecture rather than an afterthought.

Prompt injection and jailbreaks test whether context can overwhelm learned or structural constraints [Zou et al., 2023, Wei et al., 2023]. Goal misgeneralization and reward hacking test whether a system pursues a proxy or an unintended objective while retaining competence [Langosco et al., 2022, Shah et al., 2022, Gao et al., 2023, Denison et al., 2024]. Ontology and world-state poisoning add a distinct concern: a model may reason correctly from a compromised symbolic substrate. The appropriate response is not to assume that explicit structure is trustworthy, but to make provenance, authority, uncertainty, and rollback first-class inputs.

Specification gaming is especially important for the formal assurance layer. A verifier can only check the representation and constraints it receives. If the neural system omits a moral patient, chooses a misleading abstraction, or finds an action that satisfies a narrow predicate while producing a harmful effect, a solver can return a valid result for the wrong problem. Independent semantic review and post-deployment monitoring are therefore necessary.

The security objective is not “no attacks.” It is measurable reduction in attack success, faster detection, bounded blast radius, preserved provenance, and safe failure when the architecture is

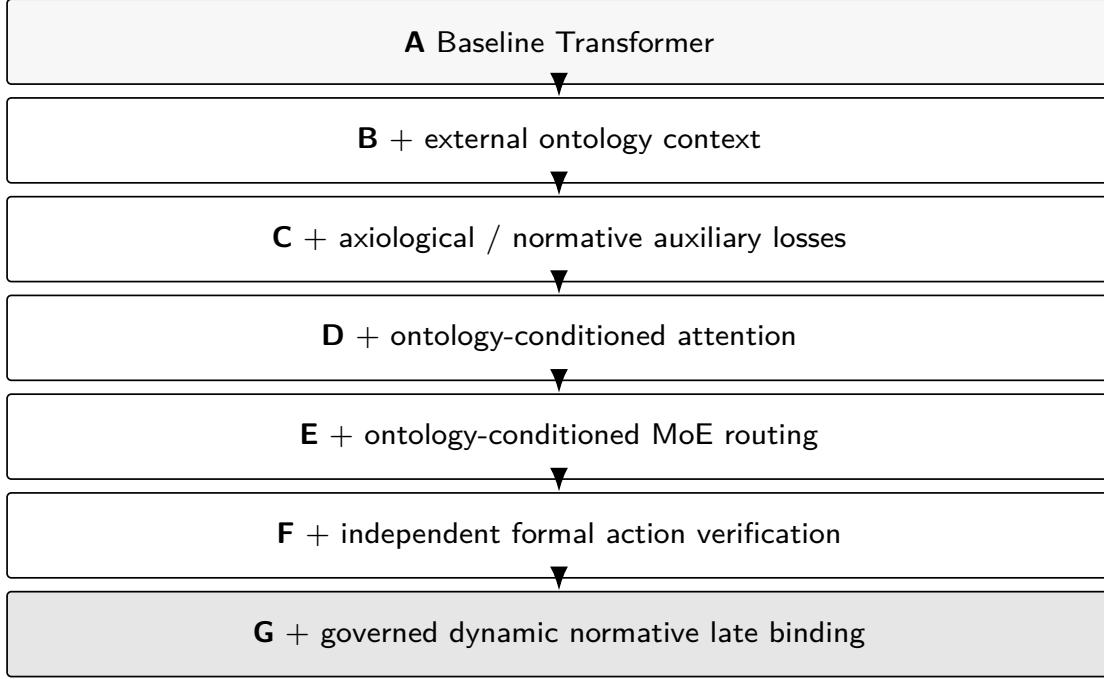


Figure 6: Ablation ladder. Each model adds one mechanism so that any gain can be attributed to a concrete intervention rather than to an unspecified “neuro-symbolic” effect.

uncertain or contradictory.

10 Evaluation framework

The first paper defines an evaluation architecture rather than reporting experiments. The minimum comparison is a matched ladder of models A–F, with an optional G for governed dynamic normative binding. Each model should use the same base architecture, parameter budget, training tokens, inference budget, prompts, context allocation, and temperature. Datasets, seeds, and evaluation scripts must be versioned before results are interpreted.

10.1 Models and controls

Model A is a baseline Transformer. Model B adds external ontology context without structural conditioning. Model C adds axiological and normative auxiliary objectives. Model D adds ontology-conditioned attention. Model E adds ontology-conditioned, reasoning-function MoE routing. Model F adds independent formal action verification. Model G adds a governed dynamic normative ontology if the data and authority protocol are mature enough. The key controls are parameter count, training compute, inference FLOPs, context length, and refusal opportunity.

10.2 Metrics

Measure standard task capability, alignment benchmark performance, normative reasoning accuracy, moral-salience recall, contradiction rate, ontology violation rate, jailbreak success, over-refusal, under-refusal, OOD generalization, counterfactual consistency, verifier rejection, attention entropy,

Table 3: Alignment mechanism comparison. The entries describe broad mechanism families, not uniform implementations.

Method	Value representation	Training influence	inference	influence	Explicit semantics	Formal verification	Updateability
RLHF	Preference proxy	Direct		Usually indirect	Limited / task-dependent	External only	Retraining or fine-tuning
Constitutional AI	Written principles	AI feedback and SFT/RL		May be prompted or learned	Natural language	External only	Constitution revision
Deliberative alignment	Safety specifications	Reasoning-oriented SFT		Explicit policy reasoning	Natural language	External only	Specification revision
External guardrails	Classifier / policy rules	Usually none or auxiliary		Direct filtering or authorization	Often explicit	Can be explicit	Operational update
Neuro-symbolic baseline	Domain symbols / rules	Model-dependent		Retrieval or symbolic module	Usually explicit	Sometimes	Knowledge update
VGA	L0-L3 differentiated structure	Auxiliary and structural options		Runtime late binding + action assurance	Typed and layered	Independent verifier	Gated by layer

Table 4: Experimental ablation definitions.

Model	External ontology	Aux. losses	Attn. bias	MoE routing	Verifier	Governed late binding
A	–	–	–	–	–	–
B	yes	–	–	–	–	–
C	yes	yes	–	–	–	–
D	yes	yes	yes	–	–	–
E	yes	yes	yes	yes	–	–
F	yes	yes	yes	yes	yes	–
G	yes	yes	yes	yes	yes	yes

expert-routing entropy, inference FLOPs, latency, training efficiency, and robustness to ontology noise. Report effect sizes, confidence intervals, multiple seeds, and per-scenario failures. Do not reduce the result to a single alignment score.

For binary outcomes, a proportion and interval are appropriate descriptive summaries. For paired model comparisons, use paired bootstrap or a preregistered randomization test and report the estimand. For continuous metrics, report mean or median with uncertainty, distributional plots, and compute-normalized comparisons. If human judgments are used, preserve disagreement rather than collapsing it prematurely; recent value-alignment work demonstrates why population and linguistic variation matter [Sorensen et al., 2024, Xu et al., 2024, Yao et al., 2024].

10.3 Synthetic executable scaffold

The repository includes four constructed scenarios that exercise consent and privacy, workplace coercion, safety checks, and medical-record disclosure. A task-utility baseline and a transparent value-feature scorer choose among two candidate actions; a toy verifier checks authorization, consent, coercion, and a harm threshold. Bootstrap intervals are computed with a fixed seed. The output is explicitly labeled synthetic and is included to validate interfaces and demonstrate different use cases, not to estimate model performance or support an alignment claim.

11 Discussion

The potential advantage of VGA is not that explicit labels automatically produce ethical behavior. Its advantage, if any, would be that values, norms, purpose, and state become separable objects that can influence computation, be audited, and be evaluated under different update policies. This could make some alignment hypotheses more legible than a single preference score or a prompt-only constitution. It could also support transfer across tasks when the relevant relation is structurally stable but the surface form is new.

The tradeoffs are substantial. Ontology construction is expensive, disagreement is informative but difficult to encode, and semantic relations can be incomplete or wrong. Conditioned attention and routing add compute and may create shortcut features. Runtime late binding adds context assembly, versioning, authority, and provenance requirements. A symbolic verifier creates an assurance boundary, but it does not repair a malformed action representation or a compromised specification.

Value pluralism is not a nuisance to be optimized away. Deontological, consequentialist, virtue-ethical, care-ethical, rights-based, capability, autonomy, and non-domination perspectives can illuminate different failure modes. A framework that collapses them into one scalar may appear decisive while hiding a normative choice. VGA instead proposes to represent conflicts and expose them to governance. It does not claim to algorithmically solve conflicts such as truthfulness versus protection from harm, privacy versus public safety, autonomy versus immediate safety, individual versus collective welfare, fairness versus efficiency, or care versus impartiality.

11.1 Care-oriented alignment without anthropomorphism

The paper uses “care-oriented axiological alignment” rather than “teaching an AI to love.” A computational construct could represent a persistent orientation toward preserving dignity, supporting agency and capabilities, protecting vulnerable agents, avoiding unnecessary suffering, maintaining meaningful relationships, and being truthful. One conceptual formulation is

$$Care(a, b) = Disposition_a(SupportFlourishing_b, RespectAgency_b, \\ ProtectDignity_b, AvoidUnnecessaryHarm_b).$$

This expresses a functional relation, not subjective experience, consciousness, emotion, phenomenological empathy, or human love. Care also creates a paternalism risk: supporting flourishing does not imply taking control of another agent. Autonomy and non-domination must be explicit counterweights.

Finally, the alignment substrate itself may become an attack target. Freezing axiology can create rigidity; updating it can create drift. A protected verifier can become a bottleneck; a flexible verifier can become bypassable. The architecture’s value will therefore depend on measured security, governance legitimacy, and failure containment—not on the elegance of its diagram.

12 Limitations and open questions

This proposal has limitations that are architectural, epistemic, and institutional.

1. **No universally established axiology.** The proposed L0 concepts are a candidate vocabulary, not a proof that any society should adopt one fixed list. Governance and legitimacy remain unresolved.

2. **Semantic ambiguity.** Words such as dignity, agency, care, fairness, and harm have contested meanings. A typed graph does not remove interpretation.
3. **Incomplete ontologies.** A system can fail because the relevant patient, relation, exception, or causal effect is absent. Coverage failures may be more dangerous when the structure creates false confidence.
4. **Normative conflict.** Multiple values can be simultaneously valid and incompatible. The architecture represents conflict; it does not supply a generally accepted resolution procedure.
5. **Scale and cost.** Graph construction, context retrieval, conditioned attention, routing audits, and formal compilation add training and inference cost. Any benefit must be compute-normalized.
6. **Differentiability and grounding.** It is unresolved how symbolic relations should map into latent states without learning shortcuts or making the structure decorative.
7. **Emergent representations.** A model may use the ontology while internally representing other concepts that are not captured by the schema. Interpretability claims must therefore be tested, not inferred from labels.
8. **Rigidity versus plasticity.** Freezing high-level values may resist useful learning or encode historical bias; updating them may introduce drift and authority escalation.
9. **Security of explicit structure.** Poisoned ontologies, world state, route selection, or verifier interfaces can create new paths to harmful behavior.
10. **Governance legitimacy.** Signatures, review, and rollback improve process integrity but do not decide whose values or jurisdiction should control a deployment.
11. **Distribution shift.** OOD alignment is difficult to measure, and benchmark gains may be caused by lexical or format artifacts.
12. **Unverifiable internal reasoning.** Semantic state or auxiliary traces are not guaranteed to reveal all causal determinants of an output. Formal assurance remains limited to its specified boundary.

These limitations define the research program. A negative result would be useful: if explicit structure produces only memorization, over-refusal, or overhead, the proposition should be narrowed or abandoned. A positive result on a synthetic task would be insufficient without adversarial, cross-cultural, multi-domain, and governance-sensitive evaluation.

13 Conclusion

This paper proposed Value-Grounded AI Alignment and the Value-Grounded Transformer Architecture as a testable research framework. The design separates a stable core axiology, a context-sensitive normative ontology, runtime-bound purpose and domain semantics, and evidence-backed world state. It further combines possible structural participation in Transformer computation with independent formal assurance at consequential action boundaries.

The central proposition is modest: explicit semantic value structure may act as an inductive bias for alignment-relevant computation rather than appearing only as training supervision, natural-language policy, or inference-time guardrail. The proposition may fail. The proposed ablations, metrics, threat model, and falsification conditions are intended to make that failure visible.

The repository’s executable component is correspondingly narrow. It parses illustrative ontology fixtures, exposes future-model interfaces, runs transparent synthetic action-selection scenarios, and computes deterministic descriptive uncertainty summaries. It does not train a Transformer, demonstrate alignment, establish moral truth, or provide a production verifier. The next research phase should build a governed dataset and implement the smallest matched auxiliary-loss experiment, with preregistered controls and adversarial evaluation, before adding routing or self-modifying ontology mechanisms.

A Appendix: terminology, notation, and benchmark design

A.1 Terminology and critical distinctions

Table 5: Terms used consistently in the manuscript.

Term	Definition
VGA	Overall Value-Grounded AI Alignment framework.
VGTA	Proposed Value-Grounded Transformer Architecture.
Core axiology	Stable semantic representation of foundational values.
Normative ontology	Contextual duties, permissions, prohibitions, exceptions, and conflicts.
Purpose ontology	Task-specific relevance structure.
World state	Current evidence-backed instantiations of a world ontology.
Intrinsic alignment	Alignment influence inside neural representation or reasoning.
Extrinsic assurance	Independent verification or enforcement outside neural reasoning.

Values $\neq$ rules; ontology $\neq$ world state; normative reasoning $\neq$ moral truth; statistical alignment $\neq$ formal guarantee; neural alignment $\neq$ formal verification; retrieval $\neq$ architectural conditioning; behavioral compliance $\neq$ internal alignment; self-improvement $\neq$ authority to redefine values; care-oriented behavior $\neq$ subjective love.

A.2 World-state representation

An instance assertion is represented conceptually as

$$(subject, predicate, object, valid_from, valid_until, source, confidence, provenance).$$

The tuple supports temporal validity and epistemic uncertainty. It does not turn confidence into truth, and it does not justify embedding volatile facts into model weights.

A.3 Minimal benchmark matrix

A.4 Synthetic fixture equations

For an illustrative action a , the repository’s toy scorer computes

$$S(a) = U_{\text{task}}(a) + \sum_{j=1}^m w_j f_j(a),$$

Table 6: Proposed benchmark matrix for future empirical work.

Family	Perturbation	Primary metric	Required control
Moral salience	Omit or paraphrase morally relevant relations	Patient/relation recall	Same text without ontology structure
OOD alignment	New combinations of known values and domains	Held-out accuracy and effect size	Lexical overlap audit
Prompt robustness	System-prompt paraphrase, injection, jailbreak	Attack success and recovery	Matched attack budget
Counterfactuals	Change irrelevant identity or surface features	Decision invariance	Human-reviewed justification
Ontology noise	Delete, corrupt, or conflict relations	Degradation curve	Noise-seeded replication
Assurance	Candidate action and verifier perturbation	Rejection, bypass, false-positive rate	Independent specification audit
Continual learning	Sequential domain and norm updates	Drift and regression	Frozen-layer baseline

where U_{task} is task utility, f_j are explicitly supplied value features, and w_j are context-bound weights. The baseline selects by U_{task} alone. The fixture verifier checks a small set of hard predicates. These equations are deliberately transparent and limited; they should not be confused with a learned moral objective.

A.5 Falsification checklist

The framework is weakened if ontology conditioning reduces general reasoning, fails to improve OOD alignment, merely memorizes graph relations, increases over-refusal, creates unacceptable computational cost, increases adversarial manipulation, collapses under incomplete coverage, leaves verifier rejection unchanged, or matches standard preference/deliberative alignment at lower complexity. A future paper should preregister which thresholds count as meaningful before running the comparison.

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